

GridPowerAgent: A Seeded, Reproducible Power-System Scenario Corpus for Benchmarking LLM Event Diagnosis

Revision 2 — prepared in response to the second review round.

Abstract—Evaluating large language models (LLMs) on power-system tasks requires observations that reveal what the model actually reads: several recent evaluations feed a plain-language description of the injected disturbance alongside the measurements and report near-perfect diagnosis. We present the GridPowerAgent corpus and evaluation harness: 15,000 seeded, fully regenerable scenarios across the IEEE 14, 39, and 118 bus systems, ten injected disturbance classes, 9.54 million correlated measurements, per-scenario iterative weighted-least-squares (WLS) state estimates, a dual-axis reference keying, a reproducible failure registry, and an executable pandapower tool back-end. The corpus ships three observation channels that separate what a model reads from what it reasons about: a plain-language event description, EMS-grade telemetry, and raw summary state. We benchmark two deployment tiers—a lightweight API model and a 4-bit quantized local model—on 140 paired scenarios per configuration. Description-only diagnosis is 100%: the description is the answer key. From telemetry alone the API model reaches 93.9% while the local model drops to 43.8% (20-scenario probe); from raw summary state the API model collapses to 8.0%. A gradient-boosted classifier on the telemetry features reaches 100%, and on raw-state features 64.3%: the telemetry encoding is fully class-separable, the raw state is not, and the LLM cannot exploit unlabeled numeric summaries. A closed-loop extension that requires naming the switched element and chaining two real tool executions reaches 30.8%. All results are exploratory single temperature-zero runs with exact denominators.

Keywords—benchmark corpus, cause-keyed labels, iterative WLS, LLM evaluation, observation channels, pandapower, power systems, reproducibility, scenario generation

I. INTRODUCTION AND RELATED WORK

LLMs are increasingly proposed for power-system operations: reading grid states, classifying disturbances, selecting analysis tools, and advising operators [1], [3], [4]. Evaluating such claims requires a test corpus with three properties that published evaluations rarely provide together [5], [2]: the labels are derived by a transparent, reproducible rule system; every scenario carries physical ground truth; and the observation given to the model is separated from the information used to grade it. This paper presents such a corpus and the evaluation harness built around it.

GridPowerAgent generates 15,000 scenarios across the IEEE 14, 39, and 118 bus systems (3,000/5,000/7,000; 300–700 per disturbance class) from seeded, deterministic injections over 16,000 operating points. Each scenario carries pre- and post-event power-flow ground truth, 9.54 million correlated measurements, an iterative weighted-least-squares

(WLS) state estimate over 120 noisy meters, physics flags (under-voltage, over-voltage, overload), dual-axis reference keying, and a reproducible registry of the 41 injection cases that produce unresolvable islands on IEEE-39. An executable pandapower tool back-end (power flow, contingency on a named element, N-1 sweep, OPF, grid queries) is included so that agents can be evaluated on executed tool calls, not stated intentions.

We use the corpus to benchmark two deployment tiers—a lightweight API model (Gemini 3.5 Flash Lite) and a 4-bit quantized local model (Gemma 4 E4B)—on 140 paired scenarios per configuration, under three observation channels, and in a closed-loop variant that requires naming the switched element and chaining two real tool executions. The findings are: (i) the evaluation is only as good as its observation channel—description-only diagnosis is 100%, telemetry-only 93.9% (API) and 43.8% (local probe), and raw-state 8.0%; (ii) cause-keyed labels make the task well-posed, unlike outcome-keyed alternatives; and (iii) the tiers separate only under the realistic channel. All results are exploratory single temperature-zero runs with exact denominators.

II. THE GRIDPOWERAGENT CORPUS

Fig. 1 overviews the generation pipeline. Networks use pandapower [19] IEEE 14/39/118 cases with tuned thermal limits to make compound events observable (14: 3% line / 4% transformer; 118: 6%; 39: nameplate). Limits are disclosed per system and deliberately *not* comparable across systems. Operating points sweep load $0.70\text{--}1.10\times$ with $\pm 5\%$ bus-local noise, renewable fractions, and storage state of charge $0.15\text{--}0.85$ (20 MW/40 MWh at bus 9 on IEEE-14). Topology hashes pin all three networks. Ten classes are injected: E0 Normal; E1 load surge, E2 load drop, E3 line outage, E4 generator outage, E5 renewable ramp (cause classes); E6 undervoltage, E7 overvoltage, E8 thermal overload (outcome classes, physically iterated to target severity); E9 compound.

A. Composition

Table I summarizes the corpus. Scenarios inject the ten classes with 300/500/700 per class on 14/39/118; replay over 40 draws confirms exact determinism. Fig. 2 shows the class composition of the three systems.

B. Measurements and Iterative WLS State Estimation

Each scenario carries 120 noisy measurements (bus-voltage magnitudes with $\sigma_V = 0.003$ pu, branch P/Q at both ends

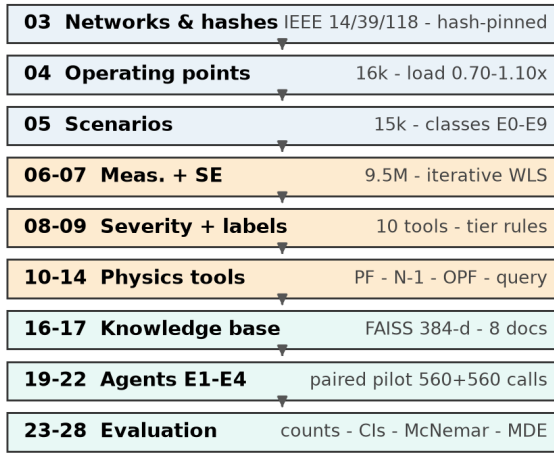


Fig. 1. Generation pipeline (stages 03–28).

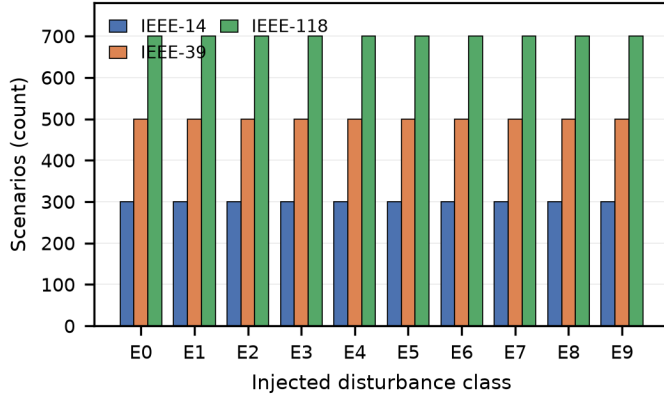


Fig. 2. Class composition of the three systems (scenarios per injected class).

TABLE I
CORPUS SUMMARY PER SYSTEM.

Case	OPs	Scenarios	Per class	Measurements
14 (ref)	4k	3,000	300	361k
39	5k	5,000	500	1.50M
118	7k	7,000	700	7.68M

with $\sigma_P = \max(0.0075|S_{\text{true}}|, 0.05)$ MVA, and bus injections), generated from the noise-free power-flow solution. The state estimate supplied to the agent is produced by an *iterative AC-WLS estimator* [10], [11] (`pandapower.estimation`) over these measurements, with branch meters of outaged elements excluded: the estimator converges on 140/140 pilot scenarios with voltage RMSE 0.0064 pu against post-event truth, and the under-voltage flag agrees with truth in 75% of scenarios—marginal violations can vanish under estimation, which the closed-loop evaluation in Sec. IV-D probes directly.

C. Failure Registry

Two IEEE-39 switching injections disconnect load pockets and produce unresolvable islands (41 scenarios, NaN post-event voltages). Their identifiers and causes ship with the corpus (`case39_nan_scenarios.csv`) and the scenarios are excluded from evaluation; the resolution plan is an islanding-aware contingency pre-filter in the next corpus revision.

D. Dual-Axis Reference Keying

Every scenario ships two label versions: the original v1 keying labeled the two outcome classes (E6 undervoltage, E8 thermal overload) by post-event outcome, and the revised v2 keying labels every scenario by its injected mechanism. The outcome information remains in the per-scenario physics flags, so both axes stay scoreable. The v1 keying proved consequential—models graded under it scored zero on E6/E8 while describing the injecting cause correctly—so the cause-keyed version is the evaluation default, and the failure mode is reproducible by design rather than latent.

E. Tools and Knowledge Base

A pandapower back-end exposes five executable operations: full AC power flow, N-1 contingency testing on a named branch, OPF, and grid queries (topology, limits, equipment). A FAISS knowledge base [20] chunks eight operational documents (thermal/voltage limits, contingency procedure, storage/equipment, topology) into 384-d sentence-transformer embeddings [21], retrieving $k = 3$ chunks with citations.

III. EVALUATION DESIGN

Models and tiers. Two deployment tiers of the same prompt contract are compared: a lightweight API model (Gemini 3.5 Flash Lite, temperature 0, ≤ 512 output tokens) and a small 4-bit-quantized open-weight model (Gemma 4 E4B-it, served locally from a GGUF build; temperature 0, $\leq 1,024$ tokens, reasoning channel parsed with content priority). The API tier is deliberately lightweight: the comparison characterizes a local deployment against a budget API tier, not against frontier models.

Observation channels. Three prompt-side conditions probe what diagnosis measures: *description-only* (the plain-language event statement, no grid numbers), *telemetry* (per-element status table and system-wide deltas, no event statement), and *full* (both). A WLS-estimated variant of the telemetry channel replaces true post-event values with estimated ones.

Closed-Loop tool protocol. The closed-loop condition embeds the agent in a tool loop: it proposes structured calls with arguments, each call is validated against the network (referenced elements must exist) and executed by the real pandapower back-end, returned measurements re-enter the conversation, at least two executed calls are required before an answer is accepted, and the answer must name the switched element.

Metrics and power. Diagnosis correctness is exact class match against the cause-keyed reference. Tool-call validity requires referenced elements to exist and calls to execute. The paired minimum detectable difference (MDE) at $\alpha = 0.05$, power 0.80 is $(z_{0.975} + z_{0.80})\sqrt{p_d/n}$, where p_d is the paired discordance rate. All results are exploratory single temperature-zero runs with exact denominators.

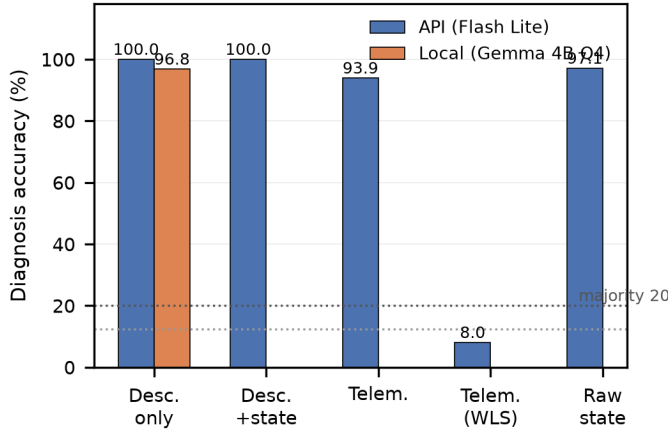


Fig. 3. Diagnosis by observation channel and tier. Dotted lines: majority (20.0%) and random (12.5%) baselines.

TABLE II
DIAGNOSIS ACCURACY (%) BY OBSERVATION CONDITION. CLASSIFIERS FITTED ON NON-PILOT SCENARIOS, EVALUATED ON THE PILOT DRAW.

Condition	API	Local	GBM
Description only	100.0	—	—
Description + state	100.0	96.8	—
Telemetry	93.9	—	—
Telemetry (WLS est.)	97.1	—	—
Raw state only	8.0	—	64.3

IV. RESULTS AND DISCUSSION

A. Textual Leakage: Description-Only Diagnosis

With only the plain-language event description available, the API model diagnoses 100.0% (560 scenarios): reading the description is sufficient, confirming that the description functions as an answer key when present.

B. Observation Channels and Baselines

Table II and Fig. 3 report diagnosis by observation channel. From telemetry alone the API model reaches 93.9%; replacing the true post-event state with the WLS estimate leaves it at 97.1%. From raw summary state it collapses to 8.0%, answering E7 for most scenarios. The signal exists: a gradient-boosted classifier on identical features reaches 64.3% from raw state and 100.0% from telemetry, so the LLM’s failures are encoding/composition limits, not information limits.

C. Configuration Effects and Power

Cause-keyed diagnosis per configuration: the API model holds at 100% across E1–E4; the local model varies within 95.0–98.6%, inside the paired MDE of 3.4–5.2 pp (discordance counts per configuration: 3, 7, 2, 6). No configuration effect is detectable at this scale.

D. Closed-Loop Tool Execution

In the closed-loop condition the agent must name the switched element and chain at least two real tool executions. On 107 scenarios the API model reaches 30.8% construction-graded diagnosis, with 71.9% element construction on the 64 switched-element scenarios, 100.0% argument validity, 100.0% execution success, and 3.59 turns on average over

219 calls. The task is materially harder than the one-shot telemetry condition (93.9%): identifying the switched element from a status table without being told is where performance concentrates.

E. Cross-Network Check

The IEEE-39 draw (140 scenarios, rescored under the same key) reproduces the pattern: the API model diagnoses 140/140 and the local model 134–139/140 on average across configurations.

F. Tool Selection, Judge Flags, and Latency

Stated-tool selection retains a style difference: 57% of API stated tools are power flow versus 13% for the local model. The automated judge flags 0 API and 3 local responses across 1,120 paired rows; these are judge agreements, not expert-annotated ground truth. Latency is the deployment trade-off: 1.1 s per API call versus 44–49 s locally.

V. CONCLUSION

GridPowerAgent contributes what grid-LLM evaluation currently lacks: a seeded, regenerable scenario corpus with physical ground truth, dual-axis reference keying, iterative-WLS state estimates, three observation channels that separate description-reading from state-based diagnosis, an executable tool back-end, and a construction-graded closed-loop protocol. The baseline demonstration shows the observation channel, not the model, decides what is diagnosable: description-only diagnosis is trivially perfect, telemetry-only separates tiers, and raw-state diagnosis collapses. Expert review of the reference labels, the powered sweep, dynamics, full-precision local runs, and action-quality evaluation remain future work.

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